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Potential impact of OPTN plan in association with donors' factors on survival times of transplanted kidneys

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ABSTRACT

The standing Organ Procurement and Transplantation Network (OPTN) committee formed in July 2003 to systematically ensure that OPTN policies and systems provide for efficient, effective, and measurable organ allocation and distribution, with the aims of (1) increasing the utility of donated organs, (2) providing for the health and safety of transplant patients, while (3) instilling public trust in the national transplant system. We investigated the potential impact of OPTN plan established in 2003.

Cox Proportional Hazard, Kaplan Meier, and ANOVA are used for survival analysis of the transplanted kidneys between 1987 and 2010 with regard to OPTN plan and donors' factors. We have observed a decrease in the proportion of failed transplanted kidneys between 2003 and 2010 in comparison with previous time intervals. The observed decrease in the proportion of the failed transplanted kidneys between 2003 and 2010 could be potentially associated with OPTN plan. The results show that the increase in the number of kidney transplants between 2003 and 2010 could also potentially be associated with the OPTN committee plan.

1. Introduction

Organ transplantation is the preferred treatment for chronic or acute organ failure and the lives of many patients have been prolonged by successful organ transplantation [1]. However, organ transplantation has become a major public health issue due to the growing disparity between organ supply and the transplantation demand. A new patient is added to the national transplant waiting list every ten minutes and twenty people die each day in the United States while waiting for a transplant. National organ donation has increased in the past two decades; yet, the number of individuals who need transplants has risen much faster. To address this disparity, the number of donors must be increased, and the discard rate of donated organs must be reduced. The discard rate for kidney has increased by almost four times from 5% to 20% between the late 1980s and 2009 [2–5].

United Network for Organ Sharing (UNOS) serves as the national Organ Procurement and Transplantation Network (OPTN) under federal contract. The total number of organ transplants performed since national transplant data were collected in 1987 has exceeded 750,000. According to UNOS, an annual record of 36,527 organ transplants performed in the US in 2018 to set the sixth straight year [6]. Although, 21,167 kidneys were transplanted in 2018 in the US, 13 people die each day while waiting for a life-saving kidney transplant. The waiting list of candidates for kidney transplantation has exceeded 100,000 [6–8]. Many patients may wait more than 5 years for an organ and they may never receive the transplant [9]. Moreover, when an organ becomes available, several factors will be considered to choose the recipient [7,8].

The US Department of Health and Human Services (HHS) joined with key national leaders in April 2003 to launch the Organ Donation Breakthrough Collaborative. The Collaborative was intended to increase access to transplantable organs. The Health Resources and Services Administration (HRSA), an agency of the HHS, launched Organ Donation Breakthrough Collaborative program in 2003 as part of the HHS Gift of Life Donation Initiative. The main goal of Organ Donation Breakthrough Collaborative program was to convince more families to donate medically eligible organs of deceased loved ones. The nationwide average rate of such donations so called the conversion rate was 52% in 2002 [10] and the program set a goal of reaching an average 75% conversion rate by 2013 [11].

UNOS is a private, non-profit organization and serves as the OPTN under contract with the HRSA [12]. The standing OPTN committee formed in July 2003 to systematically ensure that OPTN policies and systems provide for efficient, effective, and measurable organ allocation and distribution, with the aims of (1) increasing the utility of donated organs, (2) providing for the health and safety of transplant patients

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while, (3) instilling public trust in the national transplant system [13].

We have addressed the survival of transplanted kidneys based on the traveled distance and the ischemic time in our previous work [14]. We have also recently studied the graft failure time of transplanted lungs and hearts based on donors' factors [15]. In this paper, we extended our previous works and we first studied the impact of donors' factors on the survival time of the transplanted kidneys between 1987 and 2010 in the US, and second, we investigated the impact of OPTN plan on the survival time of transplanted kidneys. To achieve the former, we investigated five factors and their interactions, including donors' age, gender, history of cancer, history of cigarette smoking, and blood type of the donated kidneys. To perform the latter, we divided the dataset to three equal temporal periods. Two time periods were before and one interval was after establishment of the OPTN plan.

2. Data description

Under the federal act, UNOS is the principal non-profit association for gathering clinical information about organ donors, transplant candidates, and transplant recipients. The data for 128,992 transplanted kidneys in this study was collected from deceased donors between October 1st, 1987 and December 31st, 2010. The dataset provides the graft failure time and several factors including donors' age, gender, ethnicity, history of cancer, history of cigarette smoking, and blood type for the kidney donors.

3. Methods

To study donors' factors collectively, we performed statistical survival analysis using Kaplan Meier (KM) and Cox Proportional Hazard (Cox PH) models and analyzed the impact of donors' factors on graft failure time [16–19]. In both models, the response variable was the survival time of transplanted kidney and the independent covariate was a donor factor such as donors' age. Moreover, the relative risk was estimated to compare the survival time of transplanted kidneys for different cohorts. Additionally, ANOVA, pairwise *t*-test, and Tukey test were carried out to test the statistical hypothesis: H0: There is no significant difference in the mean survival time of transplanted kidneys in different cohorts. Finally, the interaction between these factors was investigated using ANOVA.

4. Results

We divided the transplant records to different cohorts and studied the graft failure based on the donors' factors as follows.

4.1. Age

Survival times of transplanted kidneys based on donors' factors are summarized in Table 1. We observed noticeable difference between the graft survival times depend on donors' age. Donors' age is negatively associated with the survival time, such that kidneys from younger donors are more likely to survive for longer times. The average survival time of transplanted kidneys based on age groups are 4.77, 4.32, 3.78, 3.25, and 2.36 years (with $P < 0{:}0001$).

The graft survival times are estimated using Kaplan Meier and are depicted in Fig. 1. These curves show that there is significant difference between survival times based on donors' age. The Cox proportional hazard (Tables 2 and 3) was also used to compare the risk ratio of transplanted kidneys from different age groups in comparison with the reference group (< 20). The risk ratio increases for transplanted kidneys from older donors (Table 2). The relative risks for 20 to 40, 40 to 60, 60 to 80, and greater than 80 are 1.03, 1.41, 2.06, and 3.75 respectively. Cox proportional hazard revealed that the donors' age is a significant factor regarding survival time of transplanted kidneys (Table 3).

Table 1
Descriptive statistics for kidney transplants based on donors' factors in the US from 1987 to 2010.

Donors' Factor	n	%	Mean (Std)
Age			
< 20	26,192	20.31%	4.77(3.75)
[20—40)	41,663	32.30%	4.32(3.73)
[40—60)	49,911	38.69%	3.78(3.42)
[60—80)	11,202	8.68%	3.25(3.13)
≥ 80	24	0.02%	2.36(2.33)
Gender			
Female	52,025	40.33%	4.01(3.60)
Male	76,967	59.63%	4.18(3.67)
Cigarette History			
N	84,278	65.34%	4.34(4.34)
Y	43,389	33.64%	3.99(3.58)
U	1325	1.03%	3.95(3.72)
Cancer History			
N	125,646	97.41%	4.13(3.74)
Y	3074	2.38%	3.62(3.32)
U	272	0.21%	2.64(2.76)
Blood type			
O	61,742	47.9%	4.1 (3.60)
A	19,092	14.8%	4.2 (2.90)
B	14,753	11.4%	4.0 (3.60)
AB	3368	2.6%	4.0 (3.70)

4.2. Gender

In previous research the association of gender and race with graft survival has been investigated [20,21]. The proportion of transplanted kidneys based on donors' gender is reported in Table 1. It shows that the proportion of transplanted organs from the male donors with the mean survival time of 4.18 years is about 60% and is greater than the proportion of kidneys from female donors with the mean survival time of 4.01 years. The mean survival time of transplanted kidneys from the male donors is higher than that of transplanted kidneys from the female donors (Table 1). Fig. 1 shows the Kaplan Meier survival curves of transplanted kidneys based on donors' gender. There is a significant difference between the survival curves regarding donors' gender. The Cox proportional hazard model (Tables 2 and 3) shows that there is a significant difference between the survival times of transplanted kidneys based on the donors' gender and the hazard rate of transplanted kidneys from male donors is 0.87 (95% CI: 0.86, 0.89, $P < 0.001$) and is less than the hazard rate of those from the female donors.

4.3. History of cigarette smoking

Between 1987 and 2010, about 65% of transplanted kidneys in the US were from non-smoking donors. The mean survival times of the transplanted kidneys for non-smoking and smoking donors are 4.34 and 3.99 years respectively. ANOVA shows that there is significant difference between the mean survival times of non-smoking and smoking groups for transplanted kidneys (Table 4). Kaplan Meier survival curves of the transplanted kidneys based on the donor history of cigarette smoking are demonstrated in Fig. 1. It can be seen that, there is a significant difference between the survival times of the transplanted kidneys based on the donor history of cigarette smoking (Fig. 1). The Cox proportional hazard of survival times of the transplanted kidneys (Table 3) shows that the donor smoking history has significant effect on survival times of transplanted kidneys. The risk ratio of transplanted kidneys (Table 2) from donors with cigarette smoking history is 1.17 and is 17% higher than the risk ratio of those from non-smoking donors.

4.4. History of cancer

More than 97% of transplanted kidneys between 1987 and 2010 in the US were from donors without cancer history (Table 1). The mean

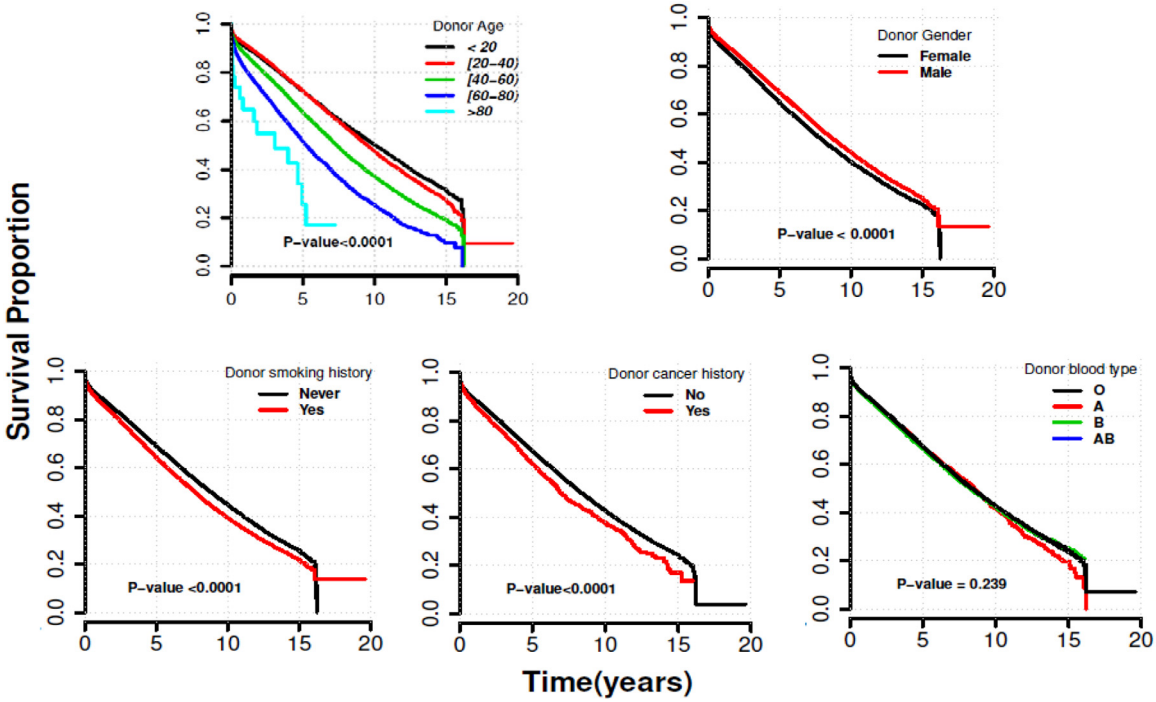


Fig. 1. Survival curves of transplanted kidneys based on donors' factors: a. Age; b. Gender; c. History of cigarette smoking; d. History of cancer; e. Blood type.

Table 2
Survival times of transplanted kidneys based on donors' factors in the US from 1987 to 2010.

Donors' Factor	RR	95% CI	*P*-value
Age			
< 20	1.00	Ref	—
[20-–40)	1.03	(1.00,1.06)	0.024*
[40-–60)	1.41	(1.38,1.45)	< 0.001
[60-–80)	2.06	(1.99,2.13)	< 0.001
≥ 80	3.75	(2.26,6.23)	< 0.001
Gender			
Female	1.00	Ref	—
Male	0.87	(0.86,0.89)	< 0.001
Cigarette History			
N	1.00	Ref	—
Y	1.17	(1.15,1.19)	< 0.001
U	1.19	(1.09,1.30)	0.000123
Cancer History			
N	1.00	Ref	—
Y	1.15	(0.92,1.45)	< 0.001
U	1.19	(1.12,1.27)	0.222
Blood type			
A	1.00	Ref	—
O	1.00	(0.98,1.02)	0.676
B	1.03	(0.99,1.06)	0.119
AB	1.02	(0.97,1.08)	0.404

survival time of the transplanted kidneys from donors with no cancer history is 4.13 years in comparison with 3.62 years for those from the donors with cancer history (Table 1). The Kaplan Meier survival curves of the transplanted kidneys based on the donor history of cancer are demonstrated in Fig. 1. Kaplan Meier survival curves show significant difference between survival times of transplanted kidneys from donors with no cancer history versus those from donors with cancer history ($P < 0.001$). Cox proportional hazard shows that hazard rate of transplanted kidneys from donors with cancer history is 15% higher than those from donors without cancer history (Table 2). Cox model shows also that cancer history is a significant factor for prediction of survival times of transplanted kidneys (Table 3). Using ANOVA, we observed significant difference between the mean survival times of

Table 3
Estimated Cox regression parameters based on donors' factors.

Donors' Factor	β	e^{β}	z	$P(Z > z)$
Age	0.013	1.013	47.38	$< 2e^{-16***}$
Gender	−0.136	0.872	−14.41	$< 2e^{-16***}$
Cigarette History				
Y	0.172	1.187	16.17	$< 2e^{-16***}$
U	0.155	1.168	3.84	0.000123***
Cancer History				
Y	0.176	1.19	5.879	< 4.1e−09***
U	0.143	1.153	1.222	0.222
Blood type				
O	−0.0042	0.995	−0.418	0.676
B	0.0247	1.025	1.56	0.119
AB	0.0218	1.022	0.835	0.404

Table 4
ANOVA test for survival times of transplanted kidneys based on donors' factors (√: Significant).

Donor Factors	Kidney
Age	√
Gender	√
Cigarette Smoking History (SH)	√
Cancer History (CH)	√
Blood type	×
Age& Gender	√
Age & SH	√
Age & CH	×
Age & ABO	×
Gender & SH	√
Gender & CH	×
Gender & ABO	×
SH & CH	×
SH & ABO	×
CH & ABO	×

transplanted kidneys based on donor history of cancer (Table 4).

4.5. Blood type

About 48% of the transplanted kidneys between 1987 and 2010 in the US were from donors with blood type O and about 15% were from donors with blood type A (Table 1). The mean survival time of the transplanted kidneys regardless of the blood type is about 4 years, while kidneys from donors with blood type A have slightly longer mean lifetime of 4.2 years (Table 1). As we can see in Table 2, Cox proportional hazard shows that kidneys from donors with blood types A and O have the same risk ratios. Kidneys from donors with blood type B (3% higher) and blood type AB (2% higher) have slightly higher risk ratios. However, the difference is not statistically significant (p-value of 0.119 for type B and p-value of 0.404 for type AB). The Kaplan Meier survival curves of the transplanted kidneys based on the donors' blood type are demonstrated in Fig. 1. Kaplan Meier survival curves does not show significant difference between survival times of transplanted kidneys from different blood types (p-value = 0.239). Cox model also shows that blood type is not a significant factor for prediction of survival times of transplanted kidneys (Table 3). Using ANOVA, we did not observe significant difference between the mean survival times of transplanted kidneys based on blood type (Table 4).

5. Failure rate with regard to organ transplantation policy

The Organ Procurement and Transplantation Network (OPTN) set up a strategy in 2003 in order to reduce the disparity between supply and demand on organs transplantation. The goals of the strategic plan were, 1. Increasing the number of transplants; 2. Reducing the organ discard rate; 3. Increasing living donation; and 4. Maximizing the use of potential donors through increased authorization rate. In order to increase donor service area (DSA), the performance of the organ procurement organizations (OPOs), critical care physicians, donor hospitals, and transplant centers must be improved [22]. Based on the OPTN plan, it was expected to have increased number of transplanted organs and reduced discard rate.

5.1. Temporal analysis of number of transplants and failed transplanted kidneys

To investigate the impacts of the OPTN plan on the number of transplanted kidneys and the proportion of the failed kidneys, we perform temporal analysis of the transplanted kidneys in the US between 1987 and 2010. We divided the data into three equal time intervals (93 months) of G1, G2, and G3 based on the transplanted date. In each interval, we calculated the number of transplanted kidneys and the number of kidneys that transplanted and failed to calculate the proportion of the failed kidneys. Moreover, we studied the number of transplanted kidneys and the proportion of failed kidneys in each time interval with regard to the donors' gender. As we can see in Table 5, the number of transplanted kidneys has increased over time from G1 to G3. To investigate whether the increase in the number of transplants has been potentially associated with the strategic policy established in 2003, we compared the number of kidney transplants in each interval with the previous interval(s). We observed steady increase in the number of kidney transplants over time. However, the increased proportion of transplanted kidneys in G3 (in comparison with G2) is considerably higher than that of G2 (in comparison with G1). Despite increased number of kidney transplants in G3, we see steady decrease in the proportion of failed kidneys.

5.2. Failed transplanted kidneys in G1, G2, and G3 with regard to donors' gender

From G1 to G3, we observed a decreasing trend in the proportion of failed kidney transplants from the male donors.% A similar decreasing trend in the proportion of failed kidney transplants from female donors was observed from G1 to G3%. The proportion of failed kidneys from male donors is less than the proportion of failed kidneys from female donors in all time intervals (G1, G2, and G3). Overall, it seems that the increased number of kidney transplants in G3 could be associated with the OPTN strategic policy. We also observed a decrease in the proportion of failed transplants in G3 in comparison with G1 and G2 in spite of increased number of kidney transplants. This could also be important and potentially associated with the OPTN strategic policy.

6. Conclusion

We studied survival time of 128,992 transplanted kidneys in this paper. We performed survival analysis, using Cox, Kaplan Meier, and ANOVA. The analysis of the survival times based on the donors' age showed a decreasing trend in the mean survival time of the transplanted kidneys by increasing donors' age. The relative risk increases for the kidneys from older donors. According to our study, the donors' gender is an important factor for transplanted kidneys. The mean survival time of transplanted kidneys from the male donors is higher than those from the female donors.

The analysis of the survival times of transplanted kidneys based on the donor history of cigarette smoking shows that survival times of kidneys from donors with smoking history is shorter than those from non-smoking donors. Donor history of cigarette smoking has a negative effect on the survival time of transplanted kidneys and the hazard ratio increases for kidneys from donors with smoking history. The donor history of cancer is a significant factor for transplanted kidneys and the hazard ratio increases for transplanted kidneys from donors with cancer

Table 5
Failure rate based on donors' factors in three-time intervals G1, G2, and G3.

Years	1987 to 1995	1995 to 2003	2003 to 2010
Organs	Kidney		
# Transplants organs	56,123	61,131	76,598
% Increased Transplants	–	8.92%	25.30%
# Failed Organs	18,088	15,414	15,090
% Failed	32.23%	25.21%	19.70%
$(p_2 - p_1) \pm \sigma$	–	(-0.070 ± 0.003)	(-0.055 ± 0.002)
# Failed from male donors (total transplants from male donors)	10,865 (35,024)	8654 (36,085)	8485 (46,024)
% Failed from male donor (total transplants from male donors)	31.02%	23.98%	18.43%
% Failed from male donor (total transplants)	19.36%	14.16%	11.08%
$(p_2 - p_1) \pm \sigma$	–	(-0.052 ± 0.002)	(-0.031 ± 0.002)
# Failed from female donors (total transplants from female donors)	7223 (21,099)	6760 (25,046)	6605 (30,574)
% Failed from female donor (total transplants from female donors)	34.23%	26.99%	21.60%
% Failed from female donor (total transplants)	12.87%	11.06%	8.62%
$(p_2 - p_3) \pm \sigma$	–	(-0.018 ± 0.002)	(-0.024 ± 0.002)

history. The analysis revealed that donors' blood type has no impact on the mean survival time of transplanted kidneys. We also investigated interaction between these factors (summarized in Table 4). For example, interaction between age and cigarette smoking history has negative impact on the survival time of transplanted kidneys. Moreover, we studied the potential impact of OPTN plan on the number of transplanted kidneys and on the proportion of failed transplants. We observed increase in the number of transplanted kidneys and decrease in the proportion of failed kidneys over time that can be associated with the OPTN plan.

It is essential to increase the survival time of transplanted kidneys to reduce the number of repeated kidney transplants. A decrease in the proportion of failed transplanted kidneys can also be interpreted as an increase in the proportion of survived kidneys. Therefore, OPTN plan can be potentially associated with increased survival rate of the transplanted kidneys. However, we should point out that optimizing the survival time of transplanted organs is a very complex procedure with several hundred collected factors from donors and recipients. The registry data is not complete and may have several missing fields for a patient. This is due to the fact that it is practically unlikely to be able to collect all factors for each patient. There is also potential interplay between recipient and donor factors with regard to organ survival. There are also other clinical developments that might concurrently be at play and affect the organ survival.

Regarding the temporal analysis and impact of OPTN, there are many other factors that possibly have changed over the time period examined. For example over past decades, there has been a substantial improvement in short term adverse outcomes such as the incidence of acute rejection related to renal transplantation. There has been better management of delayed graft function. In recent years, there have been improvements in risk/benefits of some other risk factors such as immunosuppression and could potentially affect the results. In addition, more sensitive techniques have been developed for pre-transplant immunological evaluation, which have contributed to better outcomes. Therefore, it is difficult to necessarily attribute the increase in transplants and the decrease in proportion of failed transplants to the strategic policy and a more comprehensive study is required to consider multiple factors collectively. Our future research is geared towards implementation of a more inclusive model to perform the temporal analysis.

7. Disclosure

The authors have no conflicts of interest to disclose. The data reported herein have been supplied by the United Network for Organ Sharing (UNOS) as the contractor for the Organ Procurement and Transplantation Network (OPTN). The interpretation and reporting of these data are the responsibility of the authors and in no way should be seen as official policy of or interpretation by the OPTN.

Declaration of Competing Interest

Authors do not have any conflict of interest.

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